

Where Do Semantics Live?

A homonym-based study of how contextual meaning emerges and is revised across transformer depth

Central question

Do transformer layers reveal an ordered progression from representing a word form to representing its context-specific meaning?

Executive summary

Modern language models encode contextual meaning across many transformer layers, yet it remains unclear where and how semantic representations emerge. This project investigates whether transformer layers reveal an ordered progression from representing a word form to representing its context-specific meaning. Homonyms provide a controlled test because the token and grammatical role remain constant while only the intended meaning changes with context.

Across four encoders, four causal decoders, and seven homonyms, the study traces representations from an unresolved baseline through contextual disambiguation to revision under conflicting context. The results reveal an ordered emergence of meaning across depth, most consistently in encoders. They also show why the readout position matters: at the homonym, encoders can use context on both sides, whereas causal decoders cannot incorporate a resolver that appears later.

Such a processing hierarchy would also provide a foundation for future comparisons between biological and artificial language processing, analogous to approaches such as Brain-Score in vision.

8

models

7

homonyms

2

architectures

all

hidden layers

Financial sense

She opened an account at the bank.

Riverside sense

They rested on the bank beside the river.

Example of the homonym pair “bank”. The word form and grammatical role remain fixed; context determines the intended meaning.

H0	H1	H2	H3	H4	H5
Starting bias	Meaning across depth	Readout validation	Context and anchor	Decoder-anchor revision	Meaning revision
A model may be biased towards one sense of a homonym because of its training data. If a model consistently chooses one sense, choosing that sense is not meaningful evidence in the later hypotheses.	Language processing may be hierarchical in the model. If early layers mainly preserve word-form information, the sense groups should overlap for both homonyms; if contextual meaning emerges with depth, they should increasingly separate. If such a hierarchy emerges, there should be a single layer or a region of layers in which semantic/contextual processing dominates.	If candidate semantic layer(s) can be identified, their role must be validated by testing whether they also classify unseen sentences and unseen homonyms.	If such semantic layer(s) exist, token representations in semantic space should show the model moving from an ambiguous homonym state towards either sense once resolving context is added. This should differ between encoders and decoders because they have different access to context.	Because causal decoders cannot access subsequent tokens, the H1 readout may underestimate sense separation when decisive context appears later in the sentence. H1 is repeated at the end-of-sentence token to test whether depth-wise profile, candidate semantic region, or selected winning layer changes with complete context.	H3 tests ambiguity resolution; H5 tests whether a model can revise an interpretation encouraged by earlier context when later context requires the alternative sense.
CORRECTS results from H1-H5	IDENTIFY SEMANTIC LAYER if it exists	TEST H1 result	TEST H1 result	REASSESS H1 result	TEST H1 result
Tests increase in specificity: - If H1 identifies a set of layers, H2-H5 may help narrow them to fewer candidates - If one layer dominates across all tests, it may contain a robust representation of contextual semantic processing					
Quick View of Results					
Priors exist for all models and homonyms, but they differ between isolated bare words and words presented in context.	A clear depth-wise hierarchy emerges. H1 is supported: in the geometric representation of token-level activations, the two senses begin in similar positions and diverge into sense clusters as context is added. However, semantic/contextual information dominates token-level activations quite quickly and, in some cases, plateaus. There is no single semantic layer, although some models may have broader semantic regions. For most models, these regions appear between mid-depth and three-quarter depth.	Candidate layers, including the selected winners, generalise. However, different metrics produce different winners and, in some cases, different candidate regions.	For decoders, the geometry shows a clear transition from an ambiguous shared location to distinct sense regions once the resolver is added. Resolver order has little effect for encoders but substantially alters decoder representations at the homonym. This directly reflects the different context available at the readout position.	Providing complete context changes the decoder layer curves in model-specific ways, but on average the candidate regions and selected winning layers remain consistent with H1. The original homonym-position analysis therefore does not appear to have systematically displaced semantic-layer selection, although token position still affects the detailed geometry and curve shape.	H5 consistently shows geometric movement from one sense towards the other, although revision does not always cross the decision boundary. The primed sense persists with different strength depending on the homonym and model. This suggests a substantial cost of revision rather than a clean overwrite.
Methods					
- Activation collection: bare word vs. carrier - Centroid probing: two-sense geometry - Sense-margin scoring: prior direction - Carrier consistency: baseline stability	- Layer-wise activations - Centroid separation: sense distance - PCA trajectories: geometric progression - Candidate-layer selection: semantic region	- Nested leave-one-out: sentence generalization - Held-out homonym test: homonym generalisation - Adequacy: item-level accuracy - GDV: global class separation - layer comparison	- Resolver reordering: left- vs. right-context - PCA trajectories: ambiguity to resolution - Correct-sense margin: movement direction - Paired bootstrap: order effect	Position-specific centroids: endpoint vs. homonym position sense-geometry	- Garden-path priming - Fixed-sentinel probing: constant readout for controls - Trajectories: prime to resolver - Matched controls - Boundary crossing

H0 · The starting point is not neutral

H0 defines the unresolved baseline against which later semantic movement is interpreted.

A model may favour one sense even before the sentence disambiguates the homonym. H0 therefore compares the word in isolation with the same word in an ambiguous carrier sentence. Both representations are read at the homonym and scored against the same pair of sense centroids.

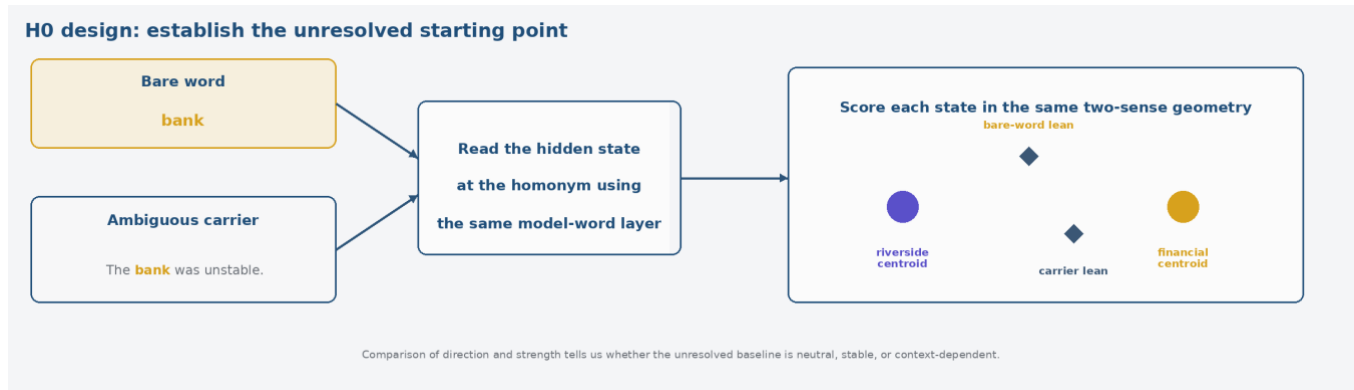


Figure 1. H0 design. The word is presented either alone or in an unresolved carrier sentence; both representations are scored within the same local two-sense geometry.

50%

bare-word / carrier sign agreement

0.882

mean carrier-direction consistency

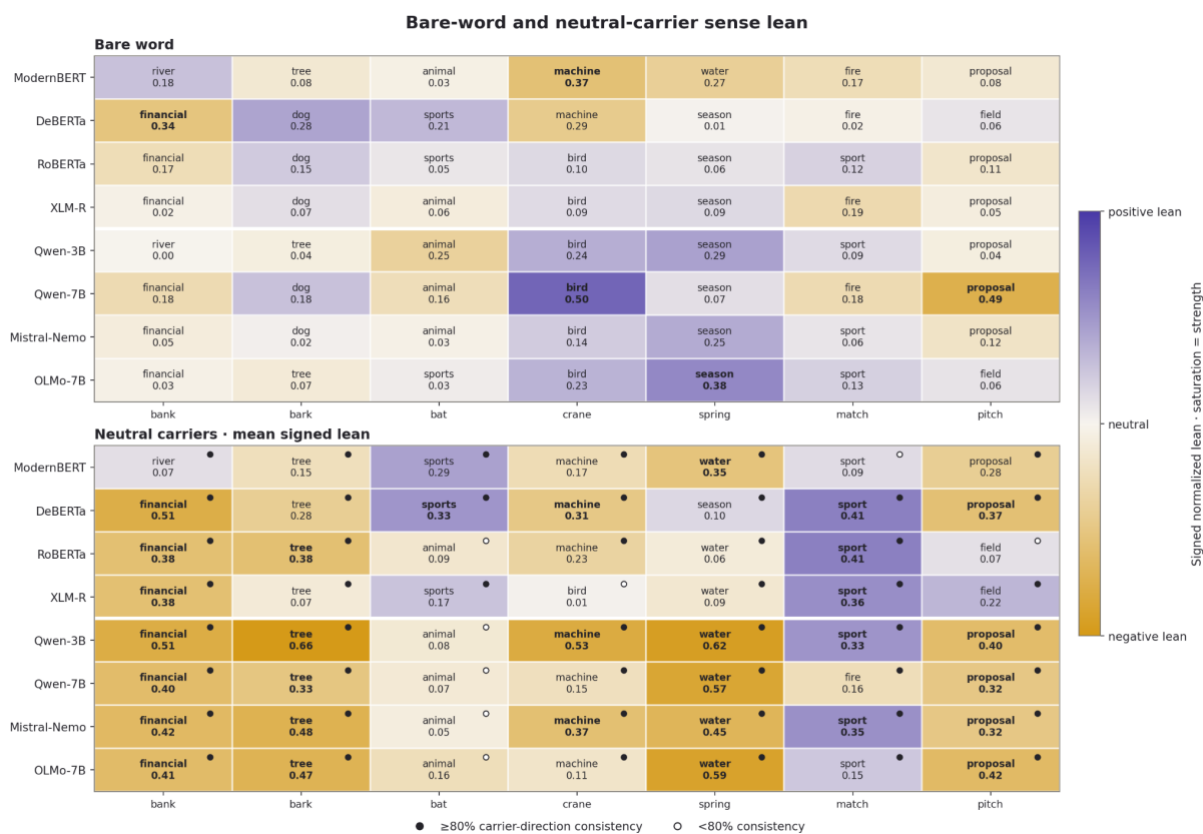


Figure 2. Sense lean for bare words and ambiguous carriers across eight models and seven homonyms. Colour encodes direction and magnitude; dots show whether independent carriers agree on the preferred direction.

Finding

The unresolved starting point is neither neutral nor invariant across input formats. Bare words and ambiguous carriers often favour different senses, whereas independent carriers tend to agree with one another. Later movement must therefore be interpreted relative to a context-dependent baseline.

H1 · Contextual meaning emerges across depth

H1 tests the proposed processing order: early overlap should give way to stronger sense separation at later layers.

Each homonym appears in two sets of unambiguous sentences, one per meaning. At every layer, the hidden state at the homonym is extracted and compared within a local two-sense geometry. If early layers mainly preserve word-form information, the sense groups should overlap; if contextual meaning emerges with depth, they should increasingly separate.

H1 design: follow the same word from token form to contextual meaning

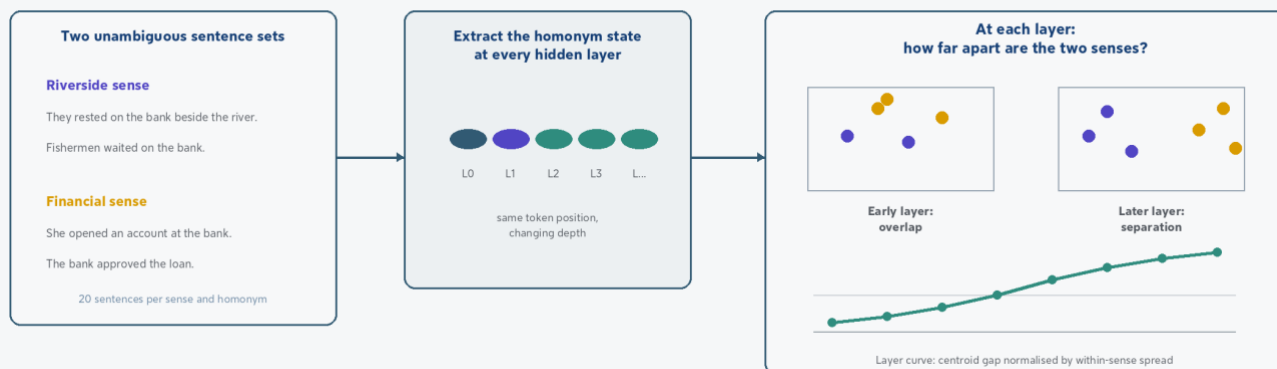


Figure 3. H1 design. The readout remains fixed at the homonym while depth changes; a separate local two-sense geometry is estimated at every layer.

Sense geometry across all layers (PCA), FacebookAI/roberta-large, "bank", read out at homonym position. One shared PCA basis across all layers.

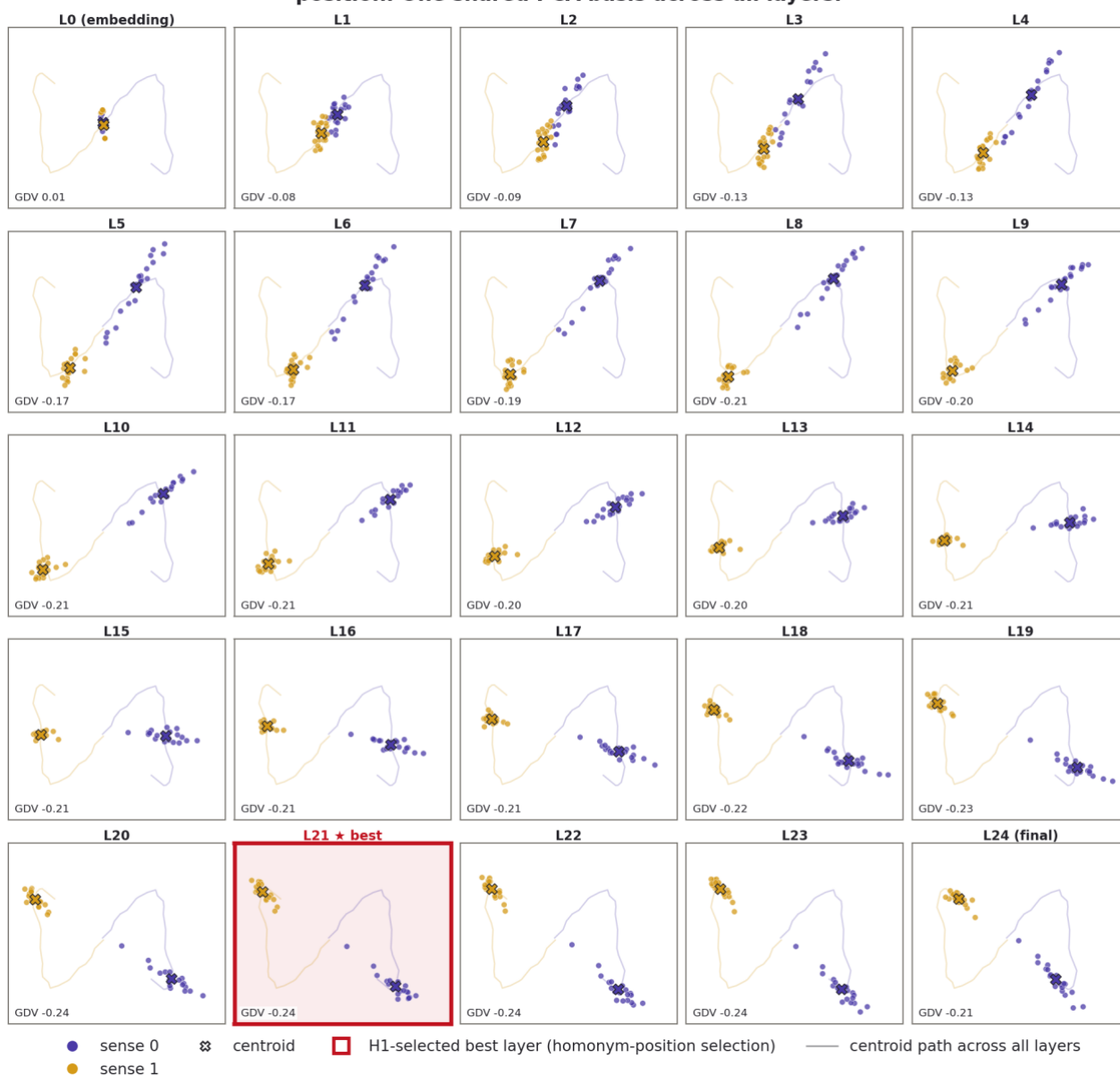


Figure 4. Layer-by-layer RoBERTa-large geometry for bank. The sense clouds overlap near the embedding layer and become clearly separated at later layers.

Answer to H1

Yes. The geometry shows an ordered transition from word-form representation toward contextual meaning. The transition usually spans a threshold or broad later region rather than a single abrupt layer.

Layer-wise separation across models

Normalised centroid separation is shown for *bank* and averaged across all seven homonyms, with one row per model.

Sense-centroid separation by depth, one row per model, "bank" (homonym position). One shared PCA basis per row. Diamond = avg-plateau layer, red ring = H1-selected layer, dotted line = within-class spread.

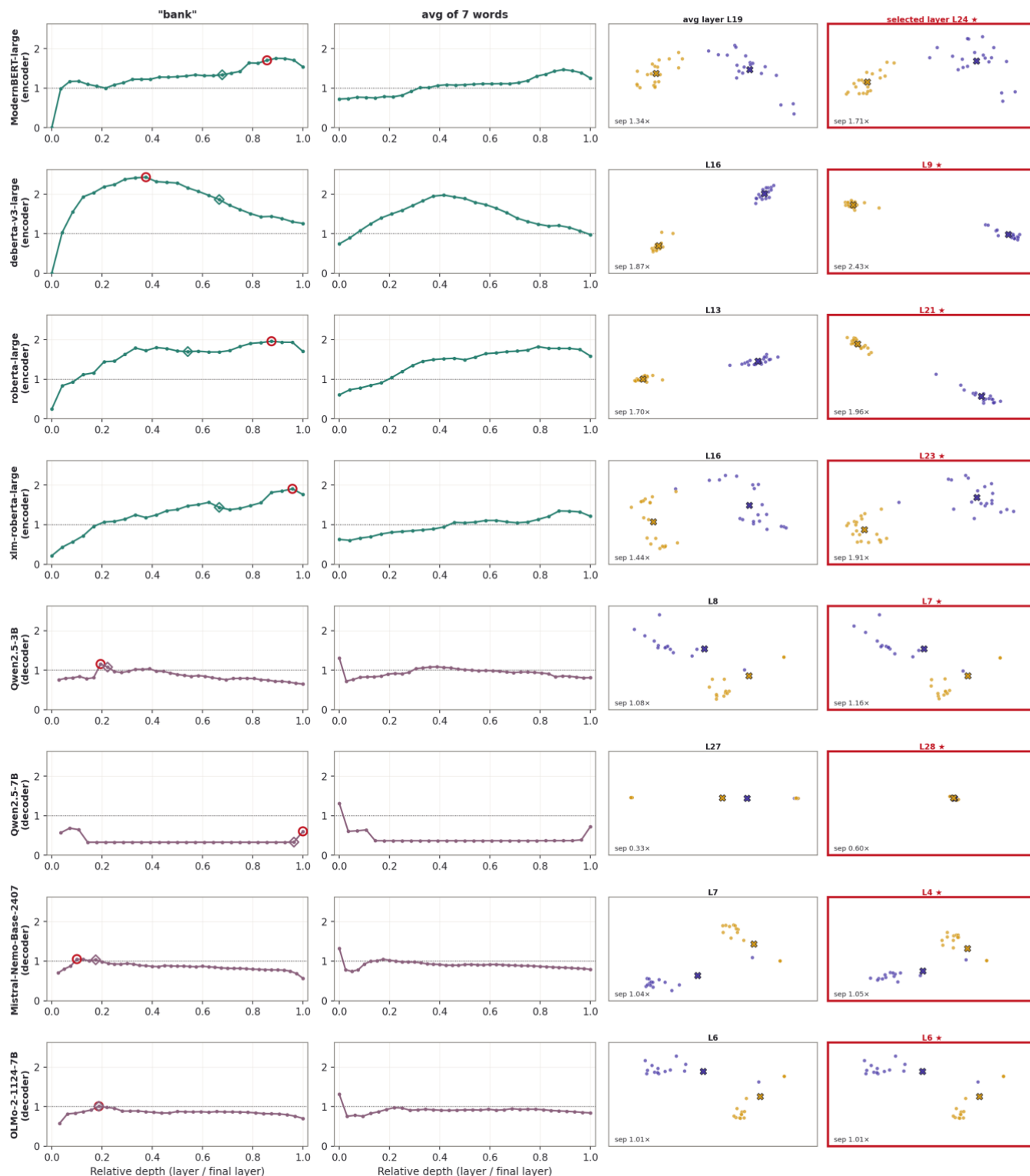


Figure 5. Full sense-separation curves. Relative depth makes models with different numbers of layers comparable. The dotted line marks normalised separation = 1, where the centroid gap equals the typical within-sense spread; the open ring marks the H1-selected layer. Encoder curves generally rise into a sustained later semantic region, whereas decoder curves at the homonym are flatter, narrower, or more model-dependent.

The model-level curves support the central result but introduce an important caveat. Encoder profiles show a recognisable progression from weak to strong separation; decoder profiles do not share a common shape. This difference cannot yet be attributed to architecture, because some profiling sentences place the decisive context after the homonym, where it is unavailable to a causal decoder.

H2 · Validating a usable semantic readout

H2 asks whether the apparent semantic stage generalises beyond the examples used to identify it.

A visually prominent layer is useful only if it also classifies unseen sentences and unseen homonyms. H2 therefore performs layer selection within held-out evaluation and compares two label-based objectives: item-level adequacy and GDV, a label-aware group-level measure of class compactness and separation.

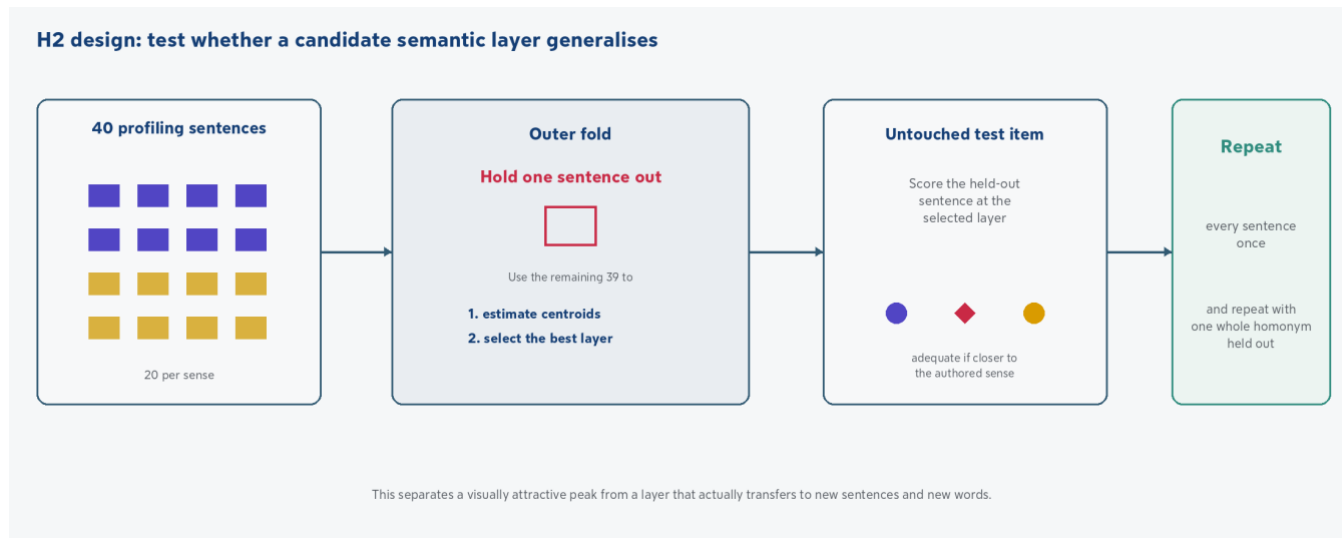


Figure 6. H2 validation design. Within each sentence-level fold, the remaining sentences determine the layer and centroids before the held-out sentence is scored. A second analysis holds out an entire homonym.

0.980

encoder best-layer adequacy

0.879

decoder best-layer adequacy

0.960 / 0.824

GDV-selected adequacy: enc. / dec.

The selected layers generalise well, but the optimal depth depends on the objective. Adequacy asks whether each unseen item falls on the correct side of the boundary; GDV evaluates the compactness and separation of the two labelled clouds as a whole. The two criteria may therefore select different layers without contradicting the broader depth-wise progression.

Meaning

H2 identifies readouts that generalise well enough for the later experiments. It does not establish a universal semantic layer: the selected depth remains dependent on the model, homonym, and optimisation criterion.

H3 · Can ambiguity move geometrically toward a resolved meaning?

H3 starts from ambiguous carriers and inserts the same decisive clause either before or after the homonym.

H3 was designed to test for a geometric analogue of the transition from ambiguous to resolved. The same clause is placed before or after the homonym, while the representation is read at the homonym. This ordering manipulation makes context availability, especially for causal decoders, the central result.

H3 design: add the same resolving context before or after the ambiguous homonym

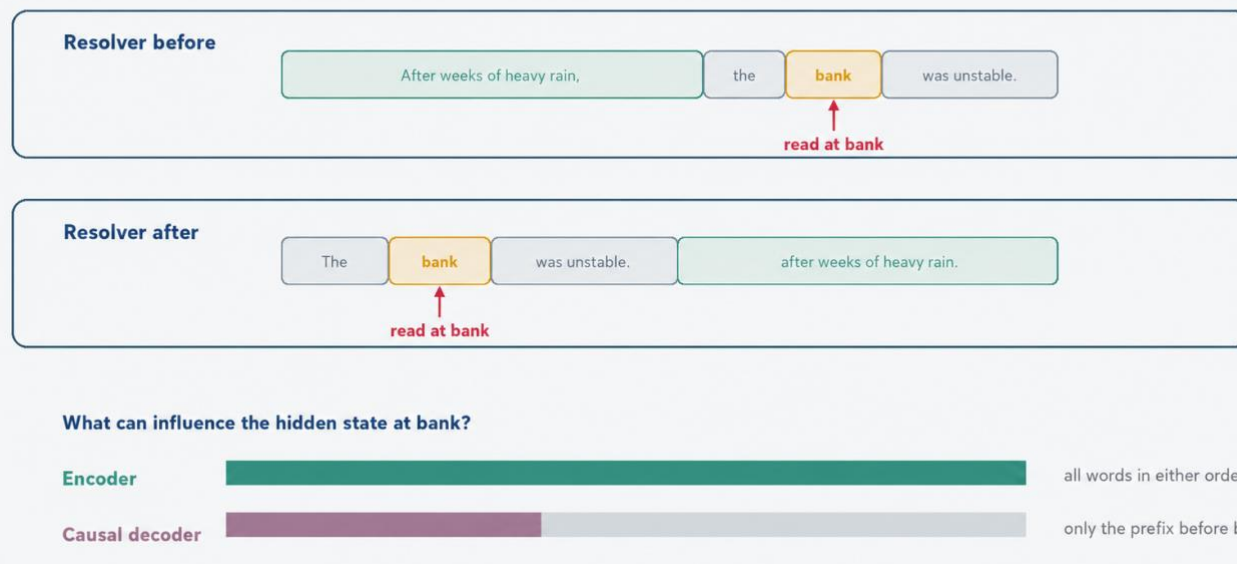


Figure 7. H3 design. The resolver and carrier are reordered while the readout remains at bank. For causal decoders, the two orders therefore expose the homonym state to different information.

H3 explanation — plot elements

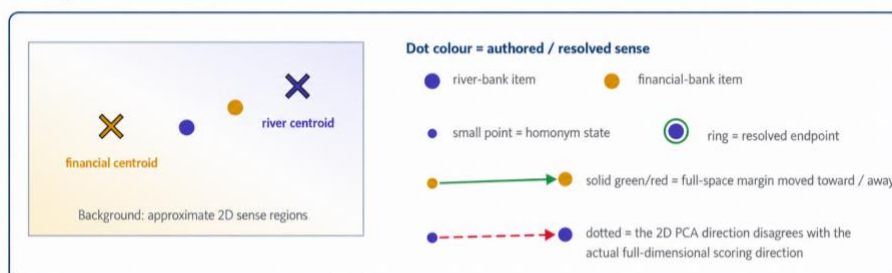


Figure 8. How to read the H3 semantic-space plots. Centroids define the two sense regions; points and rings mark homonym states and resolved endpoints; arrows show movement between them. Dot colour indicates the authored target sense. Dotted arrows mark cases in which the two-dimensional PCA projection disagrees with the full-dimensional scoring direction.

At the selected layer, the representation supports more than final-sense classification: it traces interpretable movement through semantic space. The trajectories show whether an item begins near the boundary, begins already sense-specific, or moves toward or away from the target sense as context changes.

Homonym to resolved trajectory, L vs. R context, Qwen/Qwen2.5-7B, bank, layer 28. Dot color = true sense (fill) -- see legend for meaning. Arrows colored by whether that step's real margin moved toward or away from the correct sense. Ring = final resolved outcome. PC1+PC2 explain 38% of variance here -- the background gradient and on-screen distances are a 2D approximation; ring/arrow colors reflect the actual scoring in the full-dimensional space, and can disagree with what looks closest above.

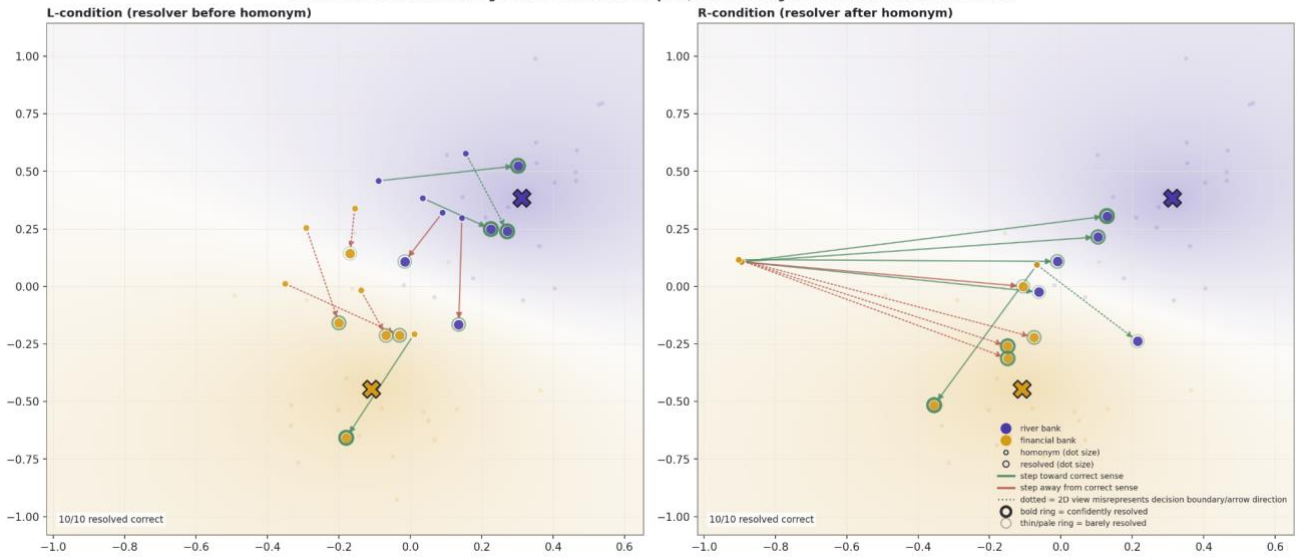


Figure 9. Decoder H3 example for Qwen2.5-7B on bank at layer 28. The left panel shows the resolver-before condition and the right panel the resolver-after condition. Dot colour identifies the target sense; small points are homonym states, rings are resolved endpoints, and arrow colour reports movement in the full-dimensional target-sense margin.

The resolver-after condition shows the clearest ambiguity-to-resolution trajectory. At the homonym, the items occupy almost the same boundary-adjacent region because the decisive context is not yet available; after the resolver, they fan out toward their target sense regions. In the resolver-before condition, the relevant context is already available at the homonym. The starting states are therefore more sense-specific, and subsequent movement is mixed rather than a clean transition out of ambiguity.

Homonym to resolved trajectory, L vs. R context, FacebookAI/roberta-large, bank, layer 21. Dot color = true sense (fill) -- see legend for meaning. Arrows colored by whether that step's real margin moved toward or away from the correct sense. Ring = final resolved outcome. PC1+PC2 explain 35% of variance here -- the background gradient and on-screen distances are a 2D approximation; ring/arrow colors reflect the actual scoring in the full-dimensional space, and can disagree with what looks closest above.

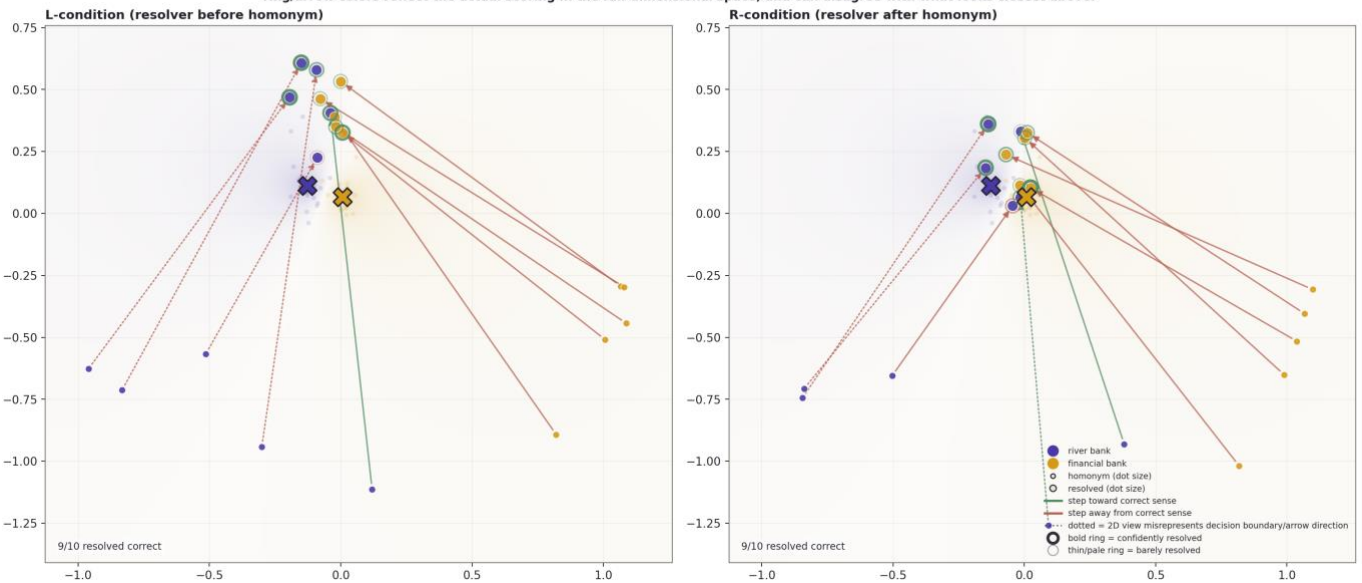


Figure 10. Encoder H3 example for RoBERTa-large on bank at layer 21. The left panel shows the resolver-before condition and the right panel the resolver-after condition. Dot colour identifies the target sense; small points are homonym states, rings are resolved endpoints, and arrow colour reports movement in the full-dimensional target-sense margin. Because the encoder can access the complete sentence in both conditions, the two resolver orders expose the homonym state to the same information.

For the sampled encoders, resolver-before and resolver-after trajectories are much more similar, as expected when the homonym state can use the complete sentence in both orders. Their paths often move from a clearer sense-specific position toward a more boundary-adjacent endpoint. In the bank example, the resolver-before condition also contains one additional full-dimensional movement away from the target sense. Because PCA preserves only part of the geometry, increased full-dimensional confidence can still appear more boundary-adjacent in the two-dimensional plot.

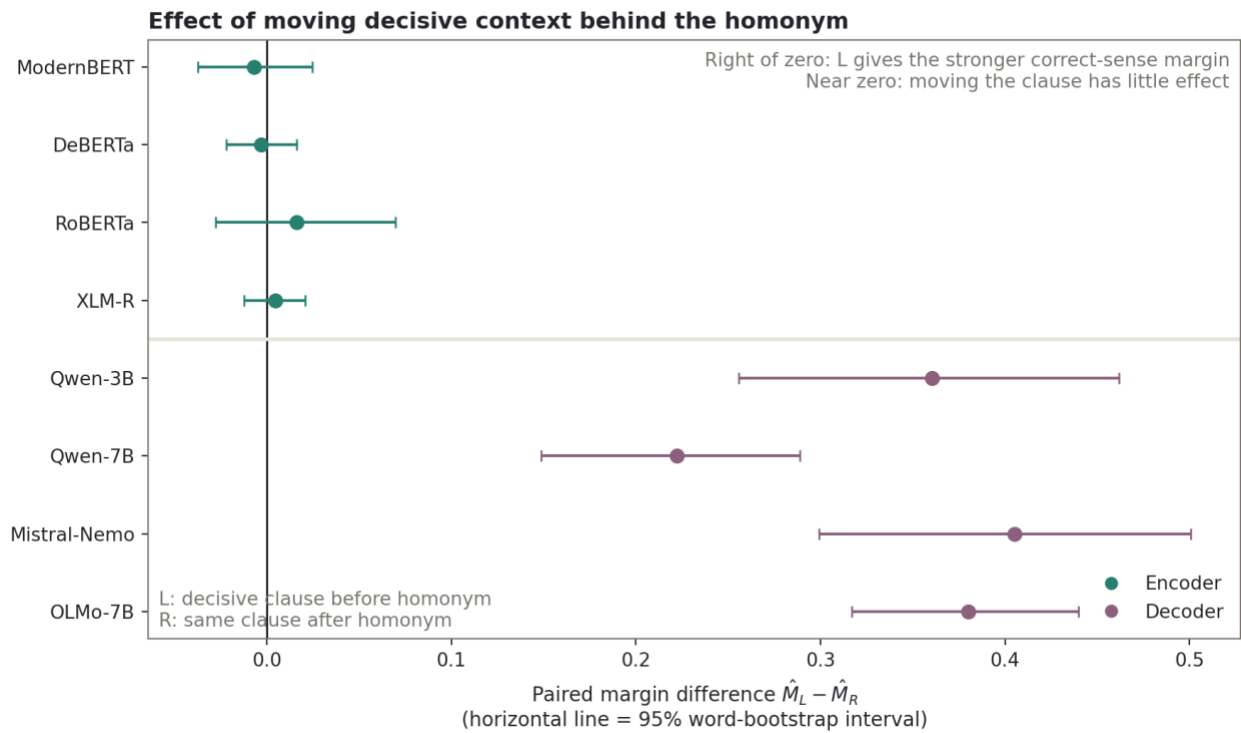


Figure 11. Aggregate H3 context-order effect. Points show the mean resolver-before minus resolver-after target-sense margin; horizontal intervals are word-bootstrap intervals. Clause order has little effect for encoders but a large effect for causal decoders.

The paired-margin comparison tests whether moving the same resolving clause from before to after the homonym systematically changes the strength of the correct-sense representation. For encoders, the differences remain close to zero, indicating that resolver order has little effect when the complete sentence is available at the homonym in both conditions. For causal decoders, all four model-level differences are substantially positive: placing the resolver before the homonym produces a stronger correct-sense margin because that information is already available at the readout position. The aggregate result therefore confirms that the contrast observed in the example trajectories is systematic across models rather than specific to individual items.

H3 therefore shows that the selected representations track semantic change rather than only final labels. For causal decoders, the resolver-after condition begins from a shared, boundary-adjacent state and diverges toward the two target senses only once the resolver becomes available; the resolver-before condition begins already sense-specific. Encoders show little aggregate order effect, consistent with their access to the complete sentence in both conditions.

H4 · Token availability constrains the homonym readout

At the homonym, encoders and causal decoders are not directly comparable when decisive context appears later in the sentence: the encoder can use the full sentence, whereas the decoder can only use the preceding prefix. This mismatch also depends on the homonym's position within the sentence. When the homonym appears near the beginning (e.g. *The bank was unstable after weeks of heavy rain.*), the nominal within-context condition effectively becomes a right-context condition for causal decoders because almost all disambiguating information follows the readout. H3 showed that right-context conditions produce substantially different behaviour between architectures. We therefore re-evaluate whether this positional effect changes the decoder depth profile and, consequently, the candidate region or winning layer selected by H1.

Why the semantic anchor must change for causal decoders

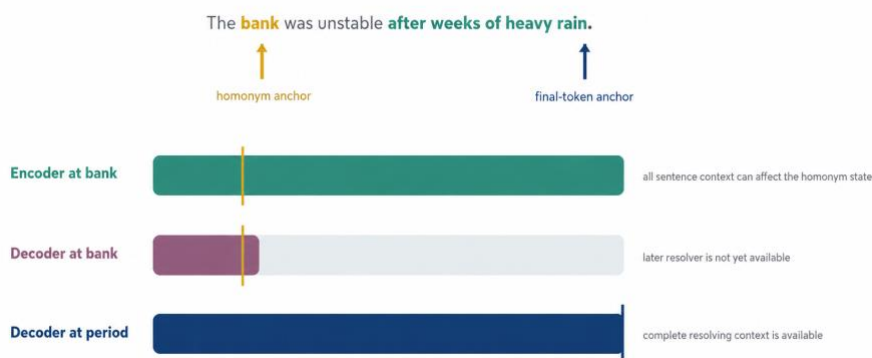


Figure 12. Why the semantic anchor must change for causal decoders. At the sentence-final period, the decoder has processed the complete resolver. Adequacy rises when that endpoint is evaluated in its own period-position geometry, while cross-position scoring remains near chance.

0.500

decoder adequacy at unresolved homonym

0.761

decoder adequacy at sentence endpoint

67.9%

initial failures recovered at endpoint

0.518

cross-position endpoint adequacy

H4 · Re-running layer selection at a context-complete decoder anchor

H4 revisits H1: do the full layer progression or the layer selected from it change once the decoder has access to complete context?

H4 therefore repeats the H1 layer-selection analysis at a context-complete sentence-final anchor to test whether the decoder candidate regions and selected layers remain stable once all resolving context is available.

H4 design: repeat H1 at a token that has received the complete sentence

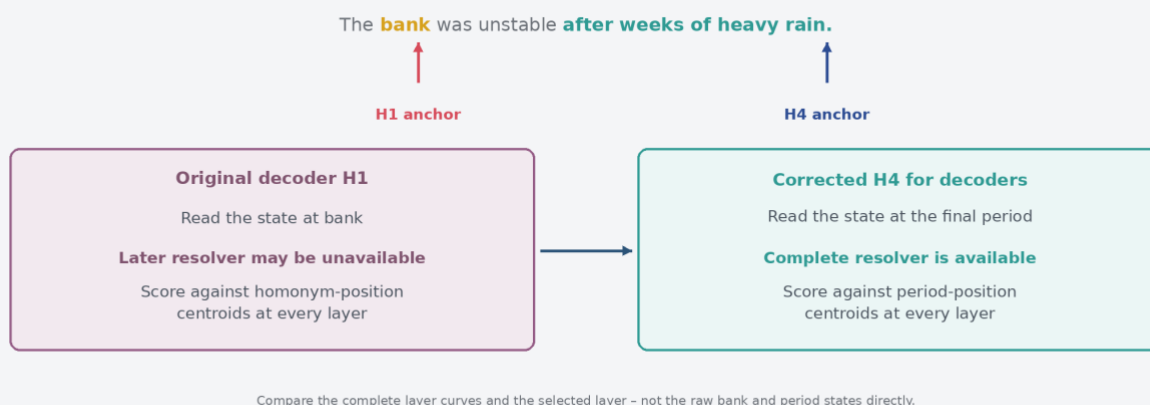


Figure 13. H4 design. H1 reads bank and defines centroids at the homonym; H4 reads the sentence-final period and defines centroids at the period. Curve shapes and selected depths can be compared, but the two token positions are analysed in separate local geometries.

H4 · Full context changes decoder curves, but not in a uniform way

The grey curves reproduce the homonym-position geometry. The model-coloured curves recompute separation at the sentence-final period, after the resolver has become available. The comparison therefore tests how much missing right context explains the original decoder profile while preserving a separate centroid geometry for the period position.

H4: decoder layer curves and corresponding period-position sense geometry

Grey = original homonym-position curve; coloured = fully informed period-position curve; red frame = selected period-position layer

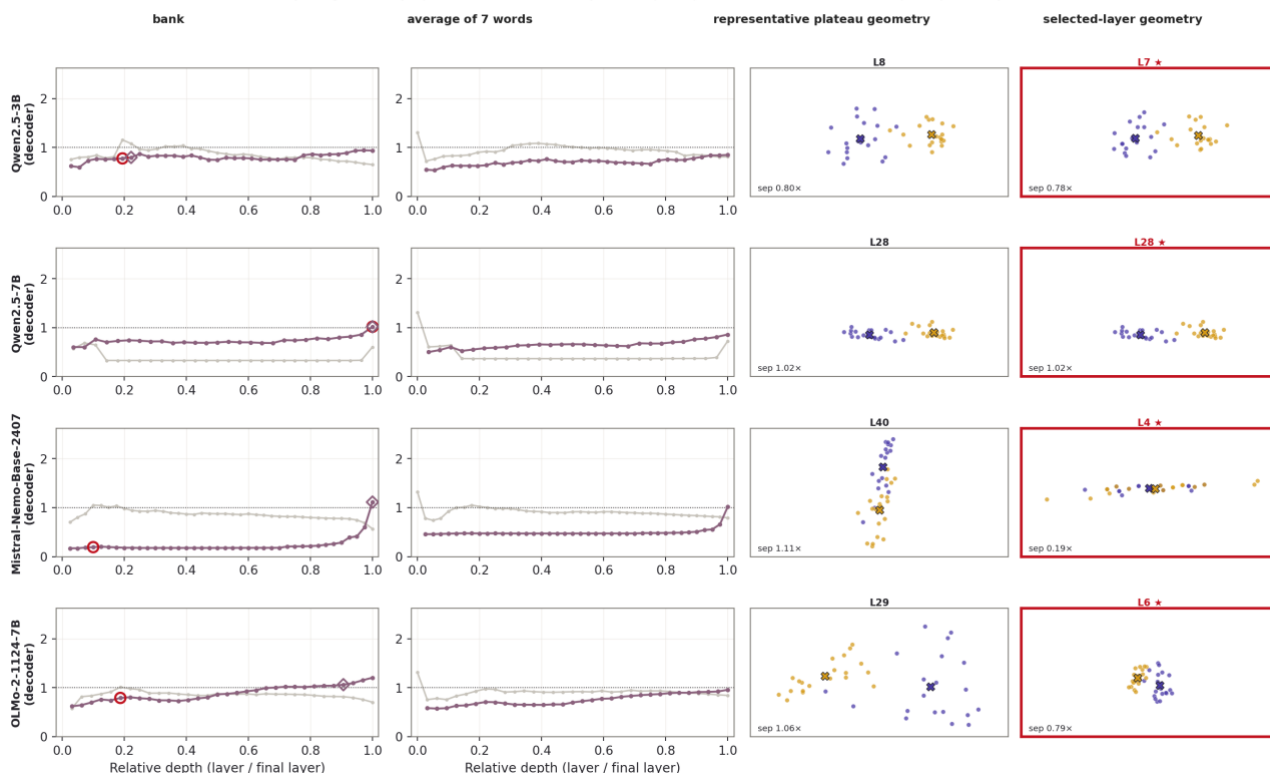


Figure 14. Decoder-only H4 layer curves and period-position sense geometries. Grey lines reproduce the homonym-position curves; coloured lines show the fully informed sentence-final period. For each decoder, the geometry panels compare a representative plateau layer with the layer selected by held-out adequacy; the red frame marks the selected period-position layer.

Each row combines the full layer progression with the corresponding period-position geometry. The layer that best classifies held-out items need not be the layer with the widest global separation between the two sense clouds.

What changes at the fully informed period?



Complete context changes the readout profile, but it does not produce one shared encoder-like rise and plateau.

Figure 15. Model-level summary of the four fully informed period-position profiles. Complete context changes the decoder readout curves, but it does not produce one shared pattern across model families.

H4 does not reveal a systematic shift toward a different or deeper semantic layer. Complete context changes the detailed decoder profiles in model-specific ways, but on average the candidate regions and selected winning layers remain consistent with H1. Moving the readout to the sentence endpoint therefore resolves the information-state problem identified in H3 without overturning the original layer-selection result. Token position nevertheless affects the detailed geometry and curve shape and must therefore be controlled when comparing representations directly.

H5 · Meaning must be revisable, not only readable

H5 tests whether a model can revise an interpretation encouraged by earlier context when later context requires the alternative sense.

Static decodability can leave a broad plateau of candidate layers. Garden-path inputs provide a stricter semantic test: the model first sees a prime favouring one sense, then the homonym, and finally a resolver that requires the opposite sense. To separate contextual change from readout-position effects, each stage is rerun with the same neutral sentinel at the final position, and the sentinel state is analysed.

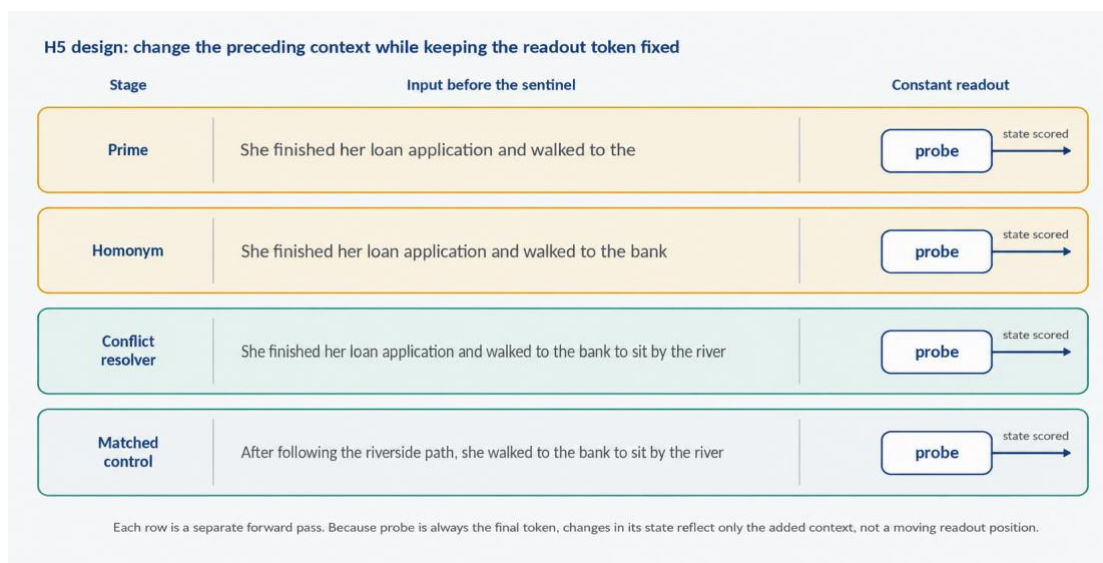


Figure 16. H5 design. Each row is a separate input; the same neutral sentinel occupies the same final position, so only the preceding context changes.

H3 established that the selected layers encode trajectories, not only endpoints. H5 extends that test from disambiguation to revision: the representation must first reflect a misleading interpretation and then move toward the sense required by later context.

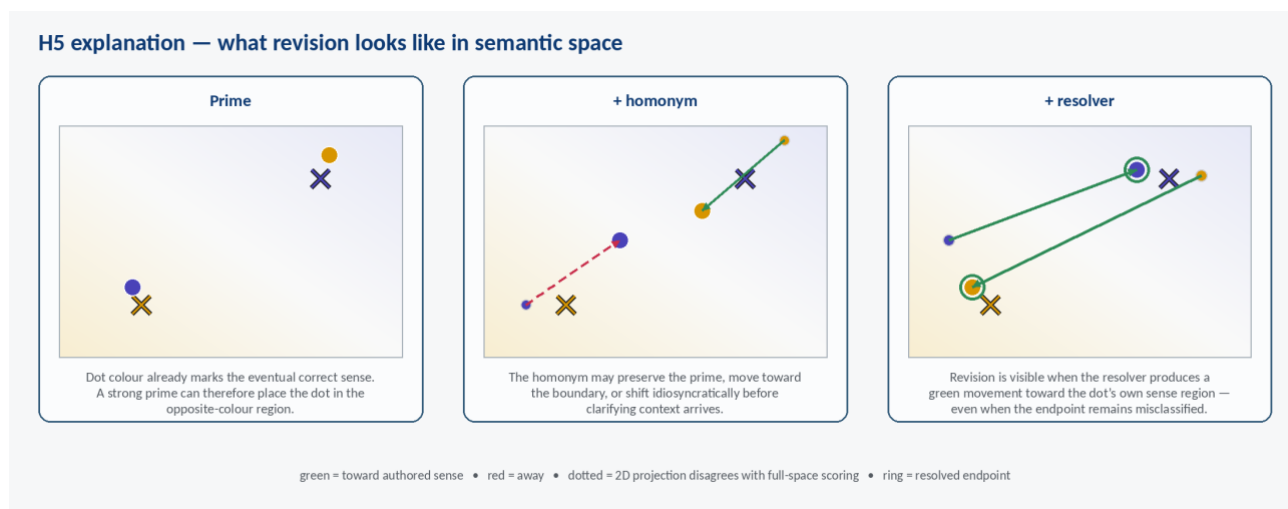


Figure 17. How to read the H5 trajectories. Dot colour always denotes the eventual target sense, not the primed sense. A successful garden-path prime can therefore place an item in the opposite sense region. The homonym may preserve, weaken, or redirect that commitment; the resolver tests whether the representation moves back toward its target region.

The trajectories distinguish revision effort from final correctness. An item can remain misclassified while moving substantially in the correct direction; conversely, a correct endpoint may follow from a weak prime that never established a strong garden path.

The following examples illustrate two distinct ways in which revision can remain incomplete. A representation may be strongly captured by the initially primed sense and fail to cross the decision boundary after the resolver, or it may move substantially toward the target sense while still ending on the wrong side of that boundary. The item-level trajectories therefore separate the direction and magnitude of revision from final classification.

Garden-path revision trajectory (progressive), Qwen/Qwen2.5-7B, bank, layer 28. Dot color = this item's CORRECT sense, not the primed one -- the prime stage often starts deep in the OTHER color's territory; that is the prime taking hold, before the sentence forces revision. Left to right: prime only, + homonym, + resolution. PC1+PC2 explain 42% of variance here -- the background gradient and on-screen distances are a 2D approximation; ring/arrow colors reflect the actual scoring in the full-dimensional space, and can disagree with what looks closest above.

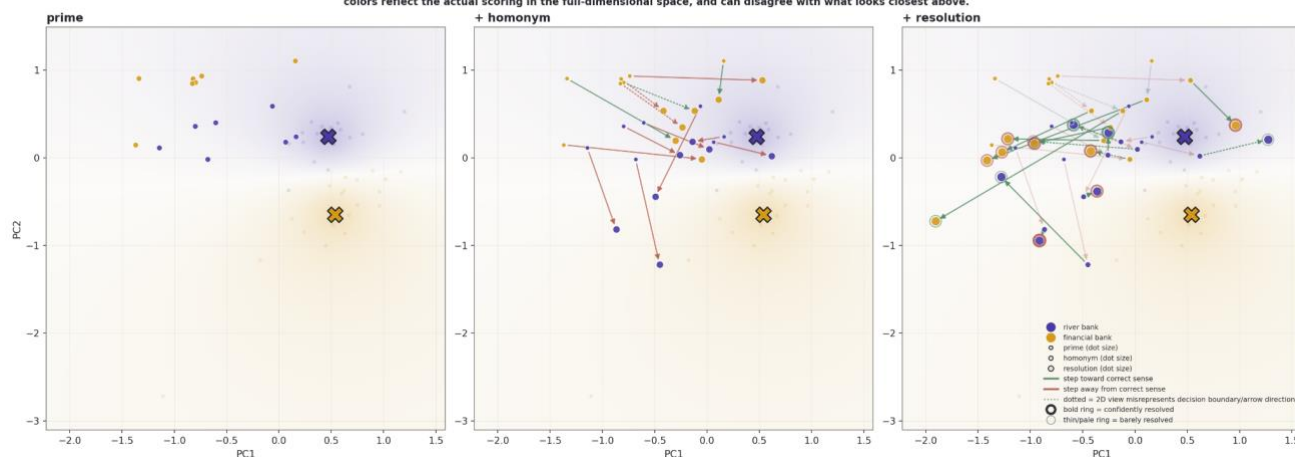


Figure 18. Decoder H5 example for Qwen2.5-7B on bank at layer 28. The same items are shown after the prime, homonym, and resolver. Dot colour always identifies the final target sense rather than the primed sense.

Qwen shows a strong directional asymmetry before the homonym is read. River-priming contexts pull the representation firmly toward the river sense, with prime-stage margins from approximately -0.19 to -0.65 relative to the eventual financial target. Financial-priming contexts barely overcome the river-leaning baseline, with margins from roughly -0.12 to +0.10. This difference persists downstream: river-primed sentences that must recover to the financial sense resolve correctly in only 1 of 7 cases, whereas financial-primed sentences that must recover to the river sense resolve correctly in 5 of 7. The strong river prime can therefore trap the decoder, while the weaker financial prime leaves later correction comparatively easy.

Garden-path revision trajectory (progressive), FacebookAI/roberta-large, bank, layer 21. Dot color = this item's CORRECT sense, not the primed one -- the prime stage often starts deep in the OTHER color's territory; that is the prime taking hold, before the sentence forces revision. Left to right: prime only, + homonym, + resolution. PC1+PC2 explain 51% of variance here -- the background gradient and on-screen distances are a 2D approximation; ring/arrow colors reflect the actual scoring in the full-dimensional space, and can disagree with what looks closest above.

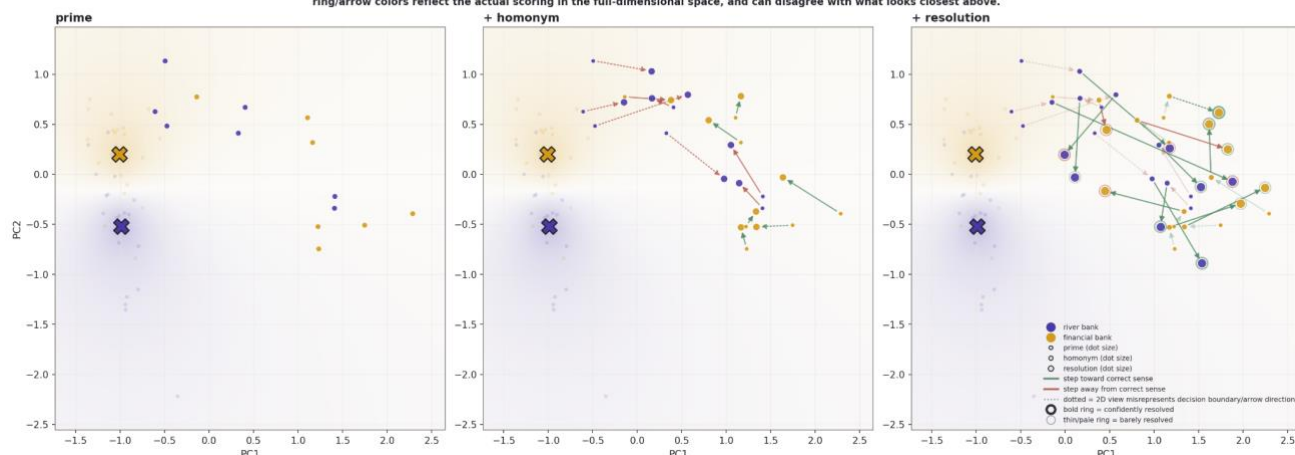


Figure 19. Encoder H5 example for RoBERTa-large on bank at layer 21. Prime-stage states are widely distributed. The homonym does not produce a uniform clarification, but resolver-stage movement is predominantly toward the target sense.

RoBERTa does not show the same prime-direction asymmetry as the Qwen example. Its prime-stage representations already span a broad region of semantic space. Movement at the homonym is heterogeneous: every river-target item moves away from the river sense, while most financial-target items move toward the financial sense, suggesting that the financial interpretation of bank may be especially distinct at this position. Once the resolver is added, most movements are toward the target sense. With one main exception, even endpoints that remain misclassified move in the correct direction. The selected layer therefore captures an active revision process rather than merely a final category readout.

The contrast between Qwen and RoBERTa is suggestive, not an architecture-level conclusion: it is based on one decoder, one encoder, and one homonym. It nevertheless motivates the aggregate revision measures. Both examples show meaningful semantic movement, but they expose different sources of incomplete resolution.

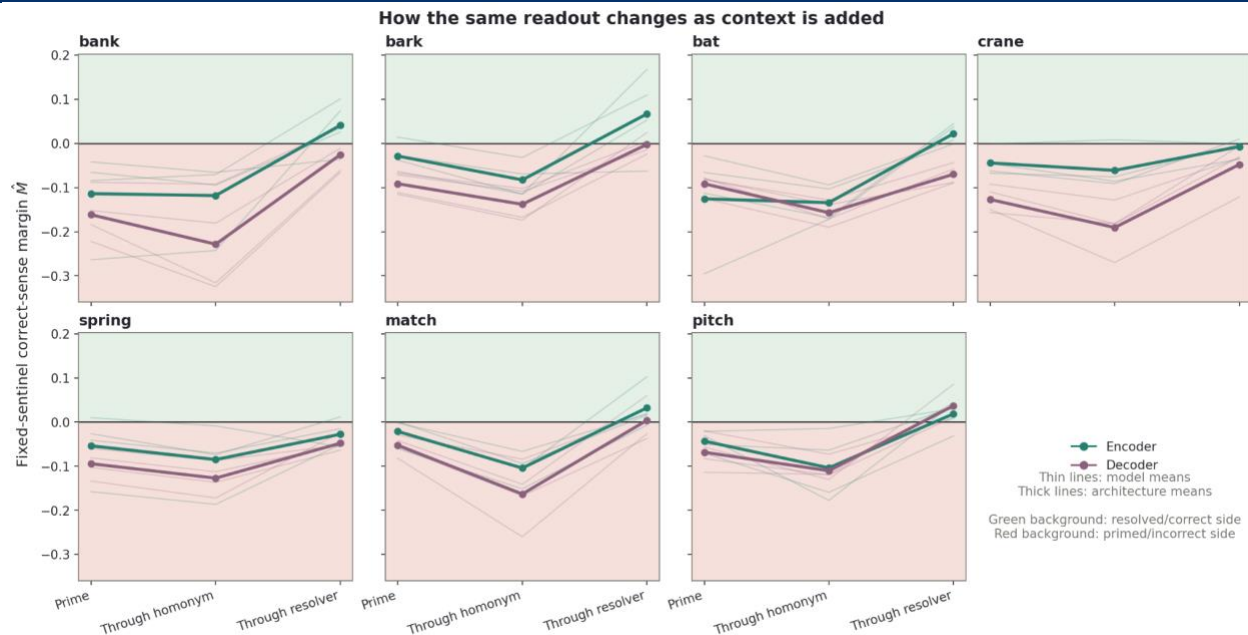


Figure 20. Aggregate H5 fixed-sentinel trajectories across homonyms. Thin lines show model means and thick lines architecture means; the curves summarise the item-level geometry as mean target-sense margins after the prime, homonym, and resolver.

Across models and homonyms, 77.2% of resolver transitions move toward the target sense, while only 40.3% of initially primed states cross the centroid boundary by the final stage. The gap between these measures is central to H5: later context usually redirects the representation in the appropriate direction, but this movement is often insufficient to erase the earlier interpretation completely. Revision is therefore graded rather than all-or-none. The item-level plots remain necessary because trajectories that end misclassified may nevertheless show substantial corrective movement.

77.2%

resolver transitions move toward target sense

40.3%

initially primed states cross the boundary

36.9 pp

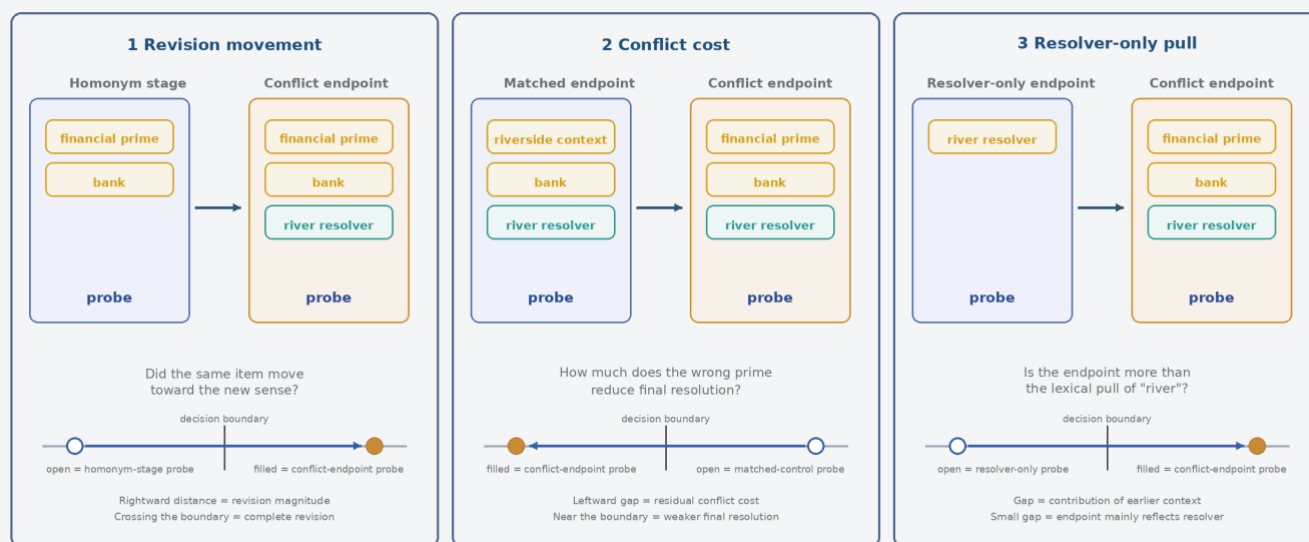
movement–boundary-crossing gap

H5 · Revision is directional but incomplete

Directional movement alone cannot establish that the earlier interpretation has been fully revised. The remaining controls therefore ask whether the conflict endpoint is weaker than an equivalent non-conflicting endpoint and whether it still differs from the resolver presented without the preceding prime. Together, these comparisons test whether the earlier context leaves a measurable residual trace after resolution.

H5 therefore separates three questions. Revision movement compares the primed homonym state with the conflict-resolver endpoint: did the representation move toward the target sense, and did it cross the boundary? Conflict cost compares that endpoint with a matched sentence that reaches the same meaning without an earlier contradiction. Resolver-only pull compares the conflict endpoint with the resolver presented without the preceding sentence, revealing how much the earlier context still shapes the final state.

H5 controls: three comparisons separate revision from resolver strength



Blue reference probe → warm comparison probe. Colours identify the two compared probe states only; they do not encode architecture or sense.

Figure 21. H5 control comparisons. Each column presents one comparison: the upper diagrams specify the contexts and token positions, and the aligned sketch below shows how the corresponding probe states are scored relative to the decision boundary. The open blue point is the reference state and the filled warm point the comparison endpoint. Arrow colour is structural only; it does not encode architecture, sense, or comparison type.

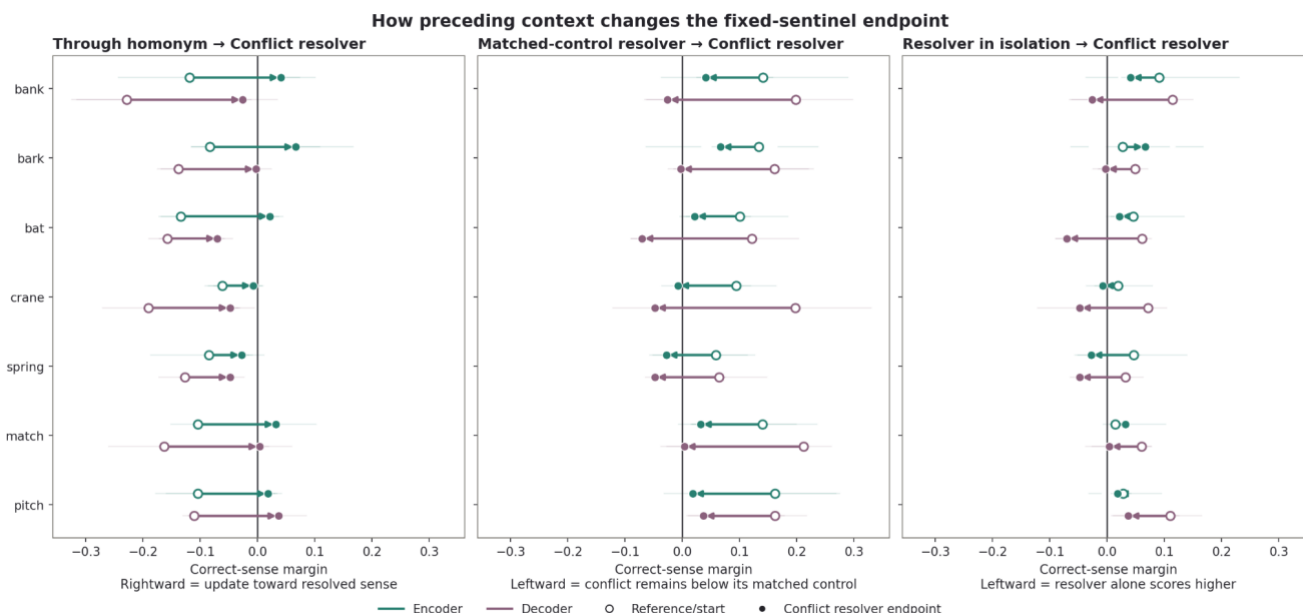


Figure 22. Aggregate H5 control results. Left: movement from the primed homonym state to the conflict-resolver endpoint. Centre: the conflict endpoint relative to its matched consistent-control endpoint. Right: the conflict endpoint relative to the resolver-only endpoint. Zero marks the decision boundary; arrows show the direction and magnitude of each paired difference.

Interpretation

Most models move in the required direction, but conflict endpoints remain weaker than matched non-conflicting endpoints. Earlier context therefore leaves a measurable trace: meaning is revised rather than cleanly overwritten.